Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Factorising Harder Quadratics

Name _____ Date _____ No calculator Version 1

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KEY WORDS factorise fully · common factor *(tauwehe pātahi)* · split the middle term · quadratic *(pūrua)*

A A common factor first ●●○

Remember: Take out the common factor, then factorise the bracket that is left.

1. Factorise fully.

a) $2x^2 + 10x + 12$

b) $3x^2 - 3x - 18$

c) $5x^2 - 20$

d) $4x^2 + 8x + 4$

e) $-x^2 + 5x - 6$

f) $x^3 + 5x^2 + 6x$

g) $6x^2 - 18x + 12$

h) $2x^2 - 50$

i) $x^3 - x^2 - 12x$

B Factorising ax² + bx + c ●●●

Remember: Find two numbers that multiply to $a \times c$ and add to b. Use them to split the middle term, then factorise in pairs.

2. Factorise.

a) $2x^2 + 7x + 3$

b) $3x^2 + 10x + 8$

c) $2x^2 - 5x + 2$

d) $5x^2 + 13x - 6$

e) $3x^2 - x - 10$

f) $4x^2 + 12x + 5$

g) $6x^2 - 7x + 2$

h) $2x^2 + x - 15$

i) $6x^2 + 5x - 6$

j) $7x^2 + 9x + 2$

k) $4x^2 - 7x + 3$

l) $10x^2 + x - 3$

C Factorising by grouping ●●●

Watch out: After the first step, both brackets must be exactly the same. If they are not, try pairing the terms differently.

3. Factorise by grouping the terms in pairs.

a) $xy + 2x + 3y + 6$

b) $ab - 4a + 5b - 20$

c) $pq + 7p - 2q - 14$

d) $mn - 6m - 3n + 18$

e) $xy - x + 4y - 4$

f) $cd + 3c - 5d - 15$

g) $ab + 2a + 8b + 16$

h) $pq - p - 7q + 7$

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D Mixed practice ●●●

4. Factorise fully. Decide which method to use first.

a) $2x^2 + 11x + 12$ **b)** $3x^2 + 21x + 30$ **c)** $25x^2 - 4$

d) $x^2 - 3x - 54$ **e)** $4x^2 + 7x - 2$ **f)** $2x^2 - 14x + 24$

g) $xy + 4x + 5y + 20$ **h)** $9x^2 - 9x - 4$ **i)** $x^3 - 8x^2 + 12x$

j) $5x^2 - 14x - 3$ **k)** $-x^2 + 9x - 20$ **l)** $49x^2 - 9$

E Simplifying fractions by factorising ●●●

> **Remember:** Factorise the top and the bottom. Then cancel any bracket that appears in both.

5. Simplify each fraction.

a) $\dfrac{x^2 + 5x + 6}{x + 2}$ **b)** $\dfrac{2x^2 + 7x + 3}{2x + 1}$

c) $\dfrac{x^2 - 9}{x^2 + 5x + 6}$ **d)** $\dfrac{x^2 + 5x + 4}{x^2 - 16}$

F Problems ●●●

6. A rectangle has an area of $2x^2 + 7x + 3$. One side is $x + 3$ long. Find an expression for the other side.

7. Hemi factorises $2x^2 + 16x + 30$ and writes $(2x + 10)(x + 3)$. Is his answer fully factorised? Explain.

★ Extension

8. Factorise $x^4 - 5x^2 + 4$ fully.

9. Factorise $2x^2 + 5xy + 2y^2$.

10. Factorise $(x + 2)^2 + 5(x + 2) + 6$. Start by treating $(x + 2)$ as a single letter.

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1-1 PRIVATE TUTORS

ALGEBRA

Factorising Harder Quadratics

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS factorise fully · common factor *(tauwehe pātahi)* · split the middle term · quadratic *(pūrua)*

A A common factor first ●●○

Remember: Take out the common factor, then factorise the bracket that is left.

1. Factorise fully.

a) $2x^2 + 10x + 8$

b) $3x^2 - 9x - 12$

c) $4x^2 - 36$

d) $5x^2 + 10x + 5$

e) $-x^2 + 6x - 5$

f) $x^3 + 6x^2 + 5x$

g) $6x^2 - 24x + 18$

h) $3x^2 - 48$

i) $x^3 - 3x^2 - 10x$

B Factorising ax² + bx + c ●●●

Remember: Find two numbers that multiply to $a \times c$ and add to b. Use them to split the middle term, then factorise in pairs.

2. Factorise.

a) $2x^2 + 5x + 3$

b) $3x^2 + 14x + 8$

c) $2x^2 - 5x + 3$

d) $5x^2 + 9x - 2$

e) $3x^2 - 5x - 12$

f) $4x^2 + 8x + 3$

g) $6x^2 - 11x + 3$

h) $2x^2 - 3x - 14$

i) $6x^2 - 5x - 4$

j) $7x^2 + 10x + 3$

k) $4x^2 - 13x + 3$

l) $10x^2 - 11x - 6$

C Factorising by grouping ●●●

Watch out: After the first step, both brackets must be exactly the same. If they are not, try pairing the terms differently.

3. Factorise by grouping the terms in pairs.

a) $xy + 5x + 2y + 10$

b) $ab - 3a + 4b - 12$

c) $pq + 6p - 3q - 18$

d) $mn - 7m - 2n + 14$

e) $xy - x + 6y - 6$

f) $cd + 5c - 4d - 20$

g) $ab + 2a + 9b + 18$

h) $pq - p - 8q + 8$

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D Mixed practice ●●●

4. Factorise fully. Decide which method to use first.

a) $2x^2 + 9x + 10$ **b)** $4x^2 + 28x + 24$ **c)** $16x^2 - 9$

d) $x^2 - 3x - 40$ **e)** $4x^2 + x - 3$ **f)** $3x^2 - 21x + 30$

g) $xy + 7x + 3y + 21$ **h)** $10x^2 + x - 2$ **i)** $x^3 - 7x^2 + 12x$

j) $5x^2 - 3x - 2$ **k)** $-x^2 + 9x - 18$ **l)** $36x^2 - 25$

E Simplifying fractions by factorising ●●●

> **Remember:** Factorise the top and the bottom. Then cancel any bracket that appears in both.

5. Simplify each fraction.

a) $\dfrac{x^2 + 7x + 12}{x + 3}$ **b)** $\dfrac{3x^2 + 7x + 2}{3x + 1}$

c) $\dfrac{x^2 - 4}{x^2 + 7x + 10}$ **d)** $\dfrac{x^2 + 7x + 10}{x^2 - 25}$

F Problems ●●●

6. A rectangle has an area of $3x^2 + 7x + 2$. One side is $x + 2$ long. Find an expression for the other side.

7. Hemi factorises $3x^2 + 27x + 42$ and writes $(3x + 21)(x + 2)$. Is his answer fully factorised? Explain.

★ Extension

8. Factorise $x^4 - 10x^2 + 9$ fully.

9. Factorise $3x^2 + 7xy + 2y^2$.

10. Factorise $(x + 2)^2 + 5(x + 2) + 4$. Start by treating $(x + 2)$ as a single letter.

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